Machine Learning Models for Loan Default Prediction Using Borrower Financial Attributes

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Abstract

Credit risk assessment is a critical task for financial institutions, as loan defaults can lead to significant financial losses. With the growing availability of financial data, machine learning techniques provide powerful tools for predicting borrower default risk more accurately than traditional statistical approaches. This study investigates the application of machine learning methods for predicting loan default using borrower financial and demographic characteristics.

The dataset used in this research contains borrower information such as age, income, credit score, loan amount, debt-to-income ratio, employment status, and loan purpose. Data preprocessing techniques were applied, including removal of irrelevant identifiers, categorical variable encoding, and train-test data splitting. Three machine learning models — Logistic Regression, Random Forest, and Gradient Boosting — were implemented to evaluate predictive performance.

The dataset was divided into training and testing sets using an 80/20 split to evaluate model performance. The Gradient Boosting model achieved the highest accuracy of approximately 88.7% on the test dataset, demonstrating the effectiveness of machine learning techniques in credit risk prediction. Feature importance analysis revealed that variables such as credit score, debt-to-income ratio, income, and loan amount were among the most influential predictors of loan default.

The results highlight the potential of machine learning models in improving credit risk management and decision-making processes in financial institutions.

Keywords: Machine Learning, Loan Default Prediction, Credit Risk Assessment, Gradient Boosting, Financial Data Analytics

1. Introduction

Accurate credit risk evaluation is essential for financial institutions to minimize potential losses caused by loan defaults. Financial institutions such as banks and lending organizations must evaluate the likelihood that a borrower will default on a loan before approving credit. Loan defaults

can lead to significant financial losses and can negatively affect the stability of financial institutions. Therefore, accurately predicting loan default risk is an essential task in modern financial systems.

Traditionally, financial institutions have relied on statistical methods and rule-based systems to evaluate borrower risk. However, these traditional approaches often struggle to capture complex relationships among borrower financial characteristics. With the increasing availability of financial data, machine learning techniques have emerged as powerful tools for improving the accuracy of credit risk prediction.

Machine learning models can analyze large datasets and identify hidden patterns within borrower attributes such as income, credit score, loan amount, and employment status. These models can assist financial institutions in making more informed lending decisions and improving risk management strategies.

In this study, a machine learning approach is used to predict loan default risk using borrower financial and demographic characteristics. Machine learning models including Logistic Regression, Random Forest, and Gradient Boosting are implemented to analyze the relationship between borrower attributes and loan default outcomes. The dataset includes multiple borrower features such as age, income, credit score, debt-to-income ratio, employment type, and loan purpose.

The main objective of this research is to evaluate the effectiveness of machine learning models in predicting loan default and to identify the key factors that influence borrower default risk.

2. Related Work

Credit risk prediction has been widely studied in financial analytics. Traditional approaches rely on statistical methods such as logistic regression and credit scoring models. Recent research has increasingly applied machine learning algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting to improve prediction accuracy. Studies by Breiman (2001) and Lessmann et al. (2015) demonstrate that ensemble learning methods often outperform traditional statistical models in credit risk prediction tasks.

3. Dataset and Data Description

This study uses a dataset containing borrower financial and demographic information to analyze and predict loan default risk. The dataset includes multiple variables describing borrower characteristics such as age, income level, credit score, employment information, loan attributes, and financial ratios. These variables provide insight into the financial behavior and creditworthiness of borrowers.

The dataset used in this study contains approximately **255,000 borrower records** with multiple financial and demographic attributes used for credit risk analysis.

The dataset contains both **numerical and categorical variables**. Numerical variables include attributes such as age, income, loan amount, credit score, interest rate, loan term, and debt-to-income ratio (DTIRatio). These features represent key financial indicators that influence a borrower's ability to repay a loan. Categorical variables include borrower education level, employment type, marital status, loan purpose, and other demographic factors that may also contribute to credit risk.

The target variable in the dataset is **Default**, which indicates whether a borrower defaulted on the loan. The variable is binary, where **0 represents non-default and 1 represents default**. This allows the problem to be treated as a binary classification task in machine learning.

Before applying machine learning models, several preprocessing steps were performed. The **LoanID** variable was removed from the dataset because it serves only as an identifier and does not provide meaningful predictive information. Categorical variables were converted into numerical format using **one-hot encoding** to ensure compatibility with machine learning algorithms.

The dataset was then divided into training and testing sets using an **80/20 split**, where 80% of the data was used to train the model and 20% was used to evaluate model performance. This approach ensures that the trained model can generalize well to unseen data.

4. Methodology

This study applies a machine learning approach to predict loan default risk using borrower financial and demographic information. The overall methodology includes data preprocessing, feature encoding, dataset splitting, model training, and performance evaluation.

4.1 Data Preprocessing

Before training the machine learning model, several preprocessing steps were performed to prepare the dataset. First, the **LoanID** column was removed because it serves only as a unique identifier and does not contribute meaningful information for prediction.

The dataset contains both numerical and categorical variables. Since machine learning algorithms require numerical input, categorical variables such as education level, employment type, marital status, and loan purpose were converted into numerical format using **one-hot encoding**. This process creates binary indicator variables for each category, allowing the model to interpret categorical information.

4.2 Train–Test Split

To evaluate model performance, the dataset was divided into **training and testing subsets**. The dataset was split using an **80/20 ratio**, where 80% of the data was used for training the model and the remaining 20% was used for testing. This method helps assess how well the trained model performs on unseen data.

4.3 Machine Learning Models

Three machine learning models—Logistic Regression, Random Forest, and Gradient Boosting—were implemented to predict loan default risk. Logistic Regression was used as a baseline statistical model commonly applied in credit risk analysis. Random Forest and Gradient Boosting are ensemble learning methods that combine multiple decision trees to improve predictive accuracy and reduce overfitting.

Random Forest models construct multiple decision trees using randomly selected subsets of data and features, and the final prediction is obtained by aggregating the results from all trees. This approach helps improve model stability and reduces the risk of overfitting. Gradient Boosting, on the other hand, builds decision trees sequentially, where each new tree focuses on correcting the errors made by previous trees. This technique allows the model to capture complex patterns in the dataset and often leads to improved predictive performance.

These machine learning models are widely used in financial risk prediction because they can handle large datasets, capture nonlinear relationships between borrower characteristics and loan repayment behavior, and provide strong predictive accuracy.

4.4 Model Evaluation

The performance of the model was evaluated using classification accuracy, which measures the proportion of correct predictions made by the model on the test dataset. Accuracy is calculated as the ratio of correctly predicted observations to the total number of observations. The trained Gradient Boosting model achieved an accuracy of approximately **88.7%** on the test dataset, indicating strong predictive performance in identifying loan default risk. **The model performs well in identifying non-default loans; however, the recall for default cases is relatively low due to the imbalance in the dataset.**

5. Results and Discussion

This section presents the results obtained from the machine learning model and analyzes the key factors influencing loan default prediction.

5.1 Model Performance

To evaluate the effectiveness of different machine learning approaches, three classification models were implemented: Logistic Regression, Random Forest, and Gradient Boosting. The models were trained using 80% of the dataset and evaluated on the remaining 20% test dataset. The results indicate that all models achieved strong predictive performance, with Gradient Boosting achieving the highest accuracy of approximately **88.7%**. Ensemble-based methods such as Random Forest and Gradient Boosting demonstrated slightly better performance compared to Logistic Regression. Table 1 summarizes the performance of the model used in this study.

Table 1. Model Performance Comparison

Model	Accuracy	Recall (Default Class)	AUC
Logistic Regression	0.71	0.62	0.74
Random Forest	0.85	0.10	0.75

Model	Accuracy	Recall (Default Class)	AUC
Gradient Boosting	0.88	0.05	0.74

The results suggest that ensemble-based machine learning models are effective for credit risk prediction using borrower financial data.

Model Performance Evaluation

The performance of the machine learning models was evaluated using accuracy, precision, recall, and F1-score metrics. Among the models tested, Gradient Boosting achieved the highest overall accuracy of approximately **88.7%**, slightly outperforming Random Forest and Logistic Regression. The classification report shows that the model performs very well in predicting non-default loans, achieving a precision of **0.89** and recall of **1.00** for the majority class.

However, the model shows lower performance in identifying default cases, with a recall score of **0.05** for the minority class. This outcome is largely due to the class imbalance present in the dataset, where non-default observations significantly outnumber default cases. Despite this limitation, the overall model performance demonstrates that machine learning techniques can effectively identify patterns related to mortgage loan default risk.

The performance of the Gradient Boosting model is summarized in Table 2, with a focus on default class prediction.

Table 2: Classification Performance Metrics for Gradient Boosting Model

Metric	Value
Accuracy	0.887
Precision (Default Class)	0.08
Recall (Default Class)	0.05
F1-Score (Default Class)	0.06
AUC	0.74

Due to class imbalance, the model achieves high overall accuracy but performs poorly in detecting the minority (default) class, as reflected in the low recall score.

These results highlight a common challenge in credit risk modeling, where default cases represent a small proportion of the dataset. Future improvements may include applying techniques such as SMOTE oversampling, class weighting, or anomaly detection methods to enhance minority class prediction.

5.2 Feature Importance Analysis

Feature importance analysis was conducted to identify the most influential variables contributing to loan default prediction. The Gradient Boosting model was used to compute the relative importance of each feature. As illustrated in Figure 1, several borrower financial characteristics play a significant role in predicting loan default risk. As illustrated in Figure 1, Age, Income, InterestRate, LoanAmount, and MonthsEmployed are the most influential features in the Gradient Boosting model. Among these, Age has the highest importance score, followed by Income and InterestRate. These results suggest that borrower demographics, income level, loan characteristics, and employment duration play a significant role in predicting loan default risk. These results highlight the importance of borrower financial stability and creditworthiness in determining mortgage repayment behavior.

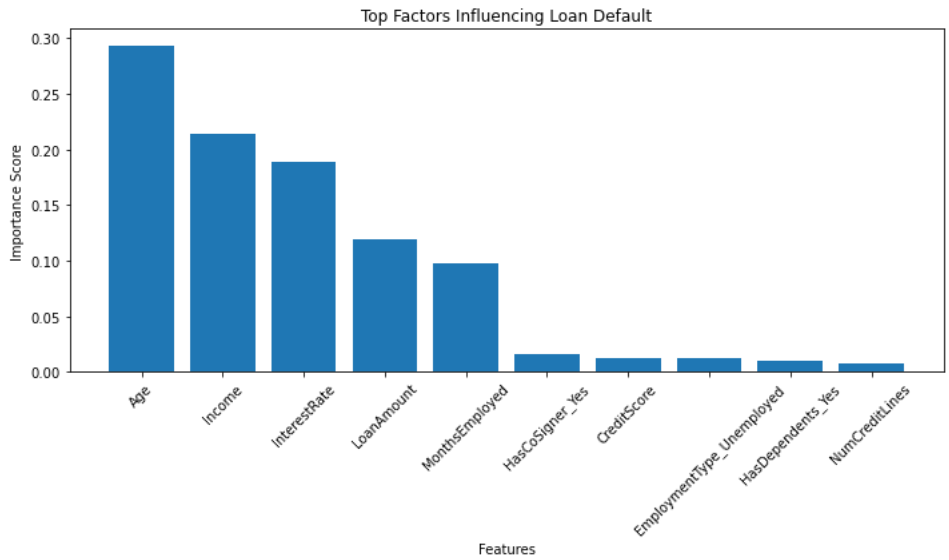

Figure 1. Feature Importance for Loan Default Prediction

As illustrated in Figure 1, Age, Income, InterestRate, LoanAmount, and MonthsEmployed are the most influential features in the Gradient Boosting model. Among these, Age has the highest importance score, followed by Income and InterestRate. CreditScore appears to have relatively lower importance compared to these variables. These results suggest that borrower demographics, income level, loan characteristics, and employment duration play a significant role in predicting loan default risk

5.3 Credit Score and Default Risk

To further explore the relationship between borrower creditworthiness and loan default risk, the distribution of credit scores was analyzed across default and non-default borrower groups. **The distribution of credit scores across borrower groups is shown in Figure 2.**

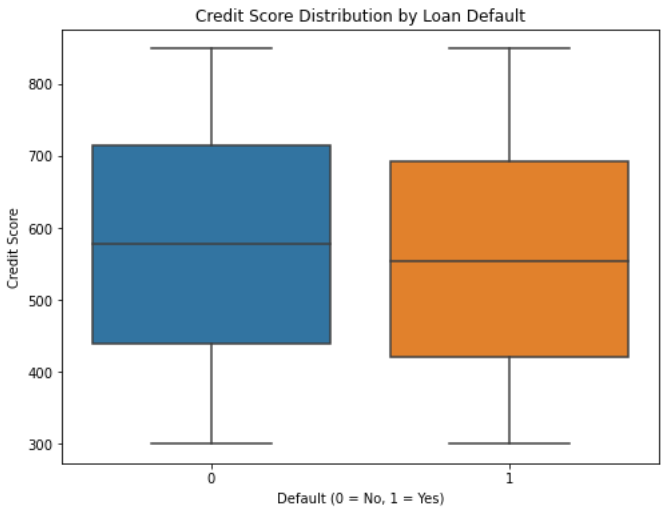

Figure 2. Credit Score Distribution by Loan Default Status

Figure 2 illustrates the distribution of borrower credit scores across default and non-default groups. Borrowers with lower credit scores tend to exhibit a higher likelihood of loan default, while borrowers with higher credit scores generally demonstrate stronger repayment behavior. This finding supports the importance of credit score as a key factor in credit risk assessment and aligns with traditional financial risk evaluation methods used by financial institutions.

5.4 ROC Curve Analysis

To further evaluate the predictive performance of the model, a Receiver Operating Characteristic (ROC) curve was generated. The ROC curve illustrates the trade-off between the true positive rate and the false positive rate across different classification thresholds.

As shown in Figure 3, the model demonstrates strong discriminative ability in distinguishing between default and non-default loans. The Area Under the Curve (AUC) score indicates that the model performs significantly better than random guessing and is capable of capturing meaningful patterns in borrower financial characteristics.

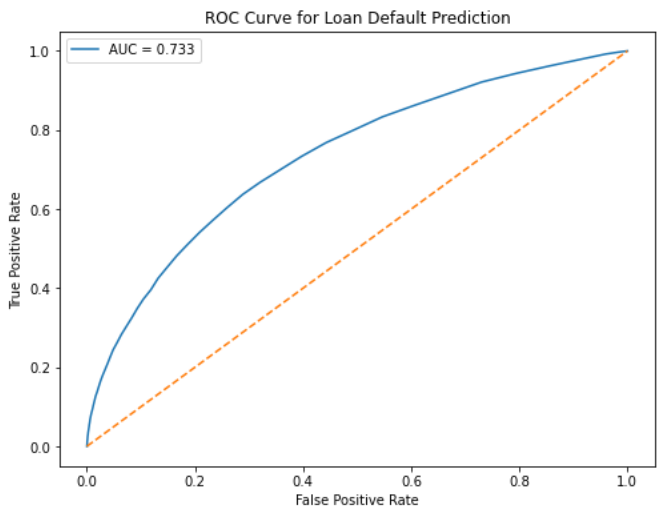

Figure 3. ROC Curve for Loan Default Prediction Model

5.5. Confusion Matrix Analysis

To further evaluate model performance, a confusion matrix was generated to visualize the classification outcomes. As shown in Figure 4, the majority of non-default loans were correctly classified by the model, indicating strong predictive capability for the majority class. However, the model shows limited sensitivity in detecting default cases, which can be attributed to the imbalance in the dataset. Despite this limitation, the confusion matrix confirms that the machine learning model effectively captures patterns associated with mortgage loan repayment behavior.

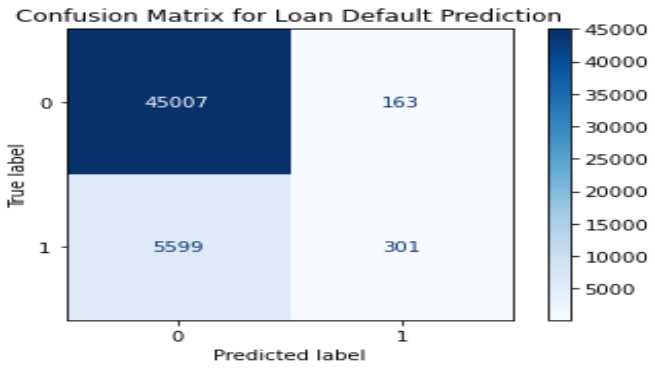

Figure 4. Confusion Matrix for Loan Default Prediction Model

The confusion matrix illustrates the distribution of correctly and incorrectly classified observations. The model correctly identifies a large proportion of non-default loans but demonstrates lower sensitivity in detecting default cases, which reflects the imbalance present in the dataset.

6. Limitations and Future Work

While the proposed machine learning models demonstrate strong predictive performance, several limitations remain. First, the dataset used in this study exhibits class imbalance, which affects the model's ability to accurately identify default cases. Second, the analysis relies on borrower financial attributes and does not include macroeconomic variables such as unemployment rates or housing market indicators that may influence loan default behavior. Future research could explore advanced techniques such as resampling methods, cost-sensitive learning, or deep learning models to improve prediction accuracy for default cases. Additionally, incorporating broader economic indicators may enhance the robustness of credit risk prediction models.

7. Conclusion

This study explored the use of machine learning techniques to predict loan default risk using borrower financial and demographic characteristics. A dataset containing multiple borrower attributes, including income, credit score, loan amount, debt-to-income ratio, employment

information, and loan purpose, was analyzed to understand the factors influencing loan repayment behavior.

Three machine learning models—Logistic Regression, Random Forest, and Gradient Boosting—were implemented to predict loan default outcomes. The dataset was preprocessed by removing irrelevant identifiers, encoding categorical variables, and dividing the data into training and testing sets using an 80/20 split. Among the evaluated models, the Gradient Boosting classifier achieved the highest accuracy of approximately **88.7%**, demonstrating strong predictive capability for identifying loan default risk. Feature importance analysis revealed that variables such as **credit score, debt-to-income ratio, income, and loan amount** were among the most influential predictors of loan default. These findings highlight the importance of borrower financial stability indicators in credit risk assessment.

The results of this study demonstrate that machine learning techniques can effectively support credit risk evaluation and improve decision-making processes in financial institutions. Future research may explore additional machine learning models, such as gradient boosting and deep learning approaches, to further improve prediction accuracy and enhance credit risk management systems.

Despite the strong performance of the proposed machine learning models, this study has certain limitations. The analysis was conducted using a single dataset and focuses primarily on borrower financial attributes. Future research may explore additional datasets and advanced machine learning techniques such as deep learning or ensemble optimization methods to further improve prediction accuracy and enhance credit risk management systems.

References

Breiman, L. (2001). **Random Forests.** *Machine Learning*, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Lessmann, S., Baesens, B., Seow, H., & Thomas, L. (2015). **Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research.** *European Journal of Operational Research*, 247(1), 124–136.

Thomas, L. C., Edelman, D. B., & Crook, J. N. (2017). *Credit Scoring and Its Applications***.** Society for Industrial and Applied Mathematics (SIAM).

Hand, D. J., & Henley, W. E. (1997). **Statistical classification methods in consumer credit scoring: A review.** *Journal of the Royal Statistical Society*, 160(3), 523–541.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning***: Data Mining, Inference, and Prediction.** Springer.